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PYMAP SCANNER

Custom Network Reconnaissance Tool

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Tool Purpose and Capabilities

PyMap is designed to accurately map network topologies, identify active nodes, and assess potential entry points via comprehensive port and service analysis.

Host Discovery

Automated ICMP Ping sweeps across CIDR ranges to map active devices on the subnet instantly.

OS Fingerprinting

Initial Time-To-Live (TTL) packet analysis to intelligently guess the underlying operating systems.

Port Scanning

Highly parallelized, multithreaded TCP Connect scans ensuring rapid and accurate port enumeration.

Service Detection

Banner grabbing execution to capture server headers, identifying precise software daemon versions.

Architecture & Libraries

PyMap is engineered entirely in Python 3, utilizing powerful standard and external libraries to handle low-level packet manipulation and concurrent processing.

- **Scapy:** Utilized for crafting raw packets and generating custom ICMP requests for host discovery without relying on system ping utilities.
- **Socket:** Implemented for establishing complete TCP handshakes and retrieving raw byte data for service banner grabbing.
- **Concurrent.futures:** Deploys a ThreadPoolExecutor with up to 50 workers to scan ports simultaneously, drastically reducing scan times.
- **Ippaddress:** Handles complex subnet mathematics, strict validation, and CIDR parsing seamlessly.



Code Review & Implementation

Modularity & Structure

The codebase is strictly separated into focused functions: `ping_sweep()`, `scan_port()`, and `grab_banner()`. This guarantees high readability, easy debugging, and simple future feature integration.

Multithreaded Execution

Sequential scanning causes massive delays due to network timeouts. Pymap resolves this by assigning individual ports (or ARP requests) to parallel threads, reducing scan times from minutes to mere seconds.

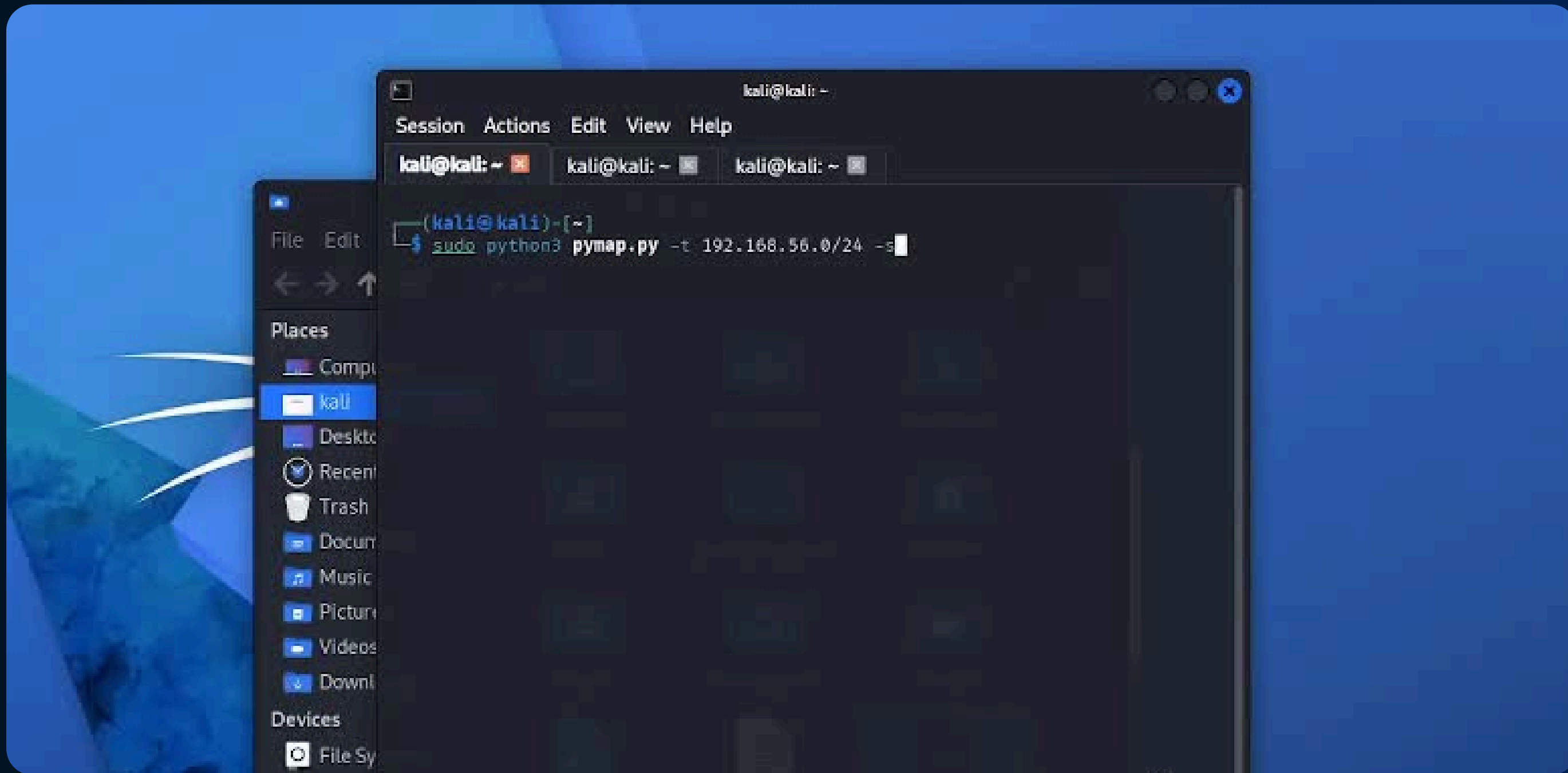
Exception & Error Handling

Comprehensive try/except blocks wrap all network operations. The script gracefully handles network timeouts, refused connections, and malformed user inputs without hard crashing.

Target Validation

Integrated checks ensure IP formats and port ranges are mathematically valid before network transmission begins, preventing wasted cycles on invalid routing attempts.

Demonstration



Limitations & Future Roadmap

Current Limitations

- **Noisy Scanning:** Full TCP Connect scans complete the 3-way handshake, meaning they are easily logged by standard IDS/IPS firewalls.
- **Basic OS Fingerprinting:** Relying purely on initial TTL values can be bypassed if network administrators modify OS default TTL settings.
- **Banner Grabbing Restraints:** Service identification relies on daemons actively broadcasting headers; stealthy services will return "Unknown".

Future Improvements

- **TCP SYN "Stealth" Scans:** Implement half-open connection scanning via Scapy to avoid triggering standard connection logs.
- **UDP Scan Integration:** Expand capabilities to detect critical stateless services like DNS and SNMP.
- **CVE Database Hooks:** Integrate an API call to dynamically cross-reference discovered banner versions against known vulnerability databases.

Lessons Learned

OSI Model Realization: Gained a tangible understanding of how packets traverse the Network Layer (ICMP protocol creation) and Transport Layer (TCP 3-way handshake).

The Power of Concurrency: Learned firsthand that sequential network operations are fundamentally flawed due to physical latency, making multithreading a mandatory skill.

Demystifying Cyber Tools: Stripped away the "magic" of enterprise scanners like Nmap, revealing that they operate on standard, predictable networking protocols and socket interactions.

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THANK YOU

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